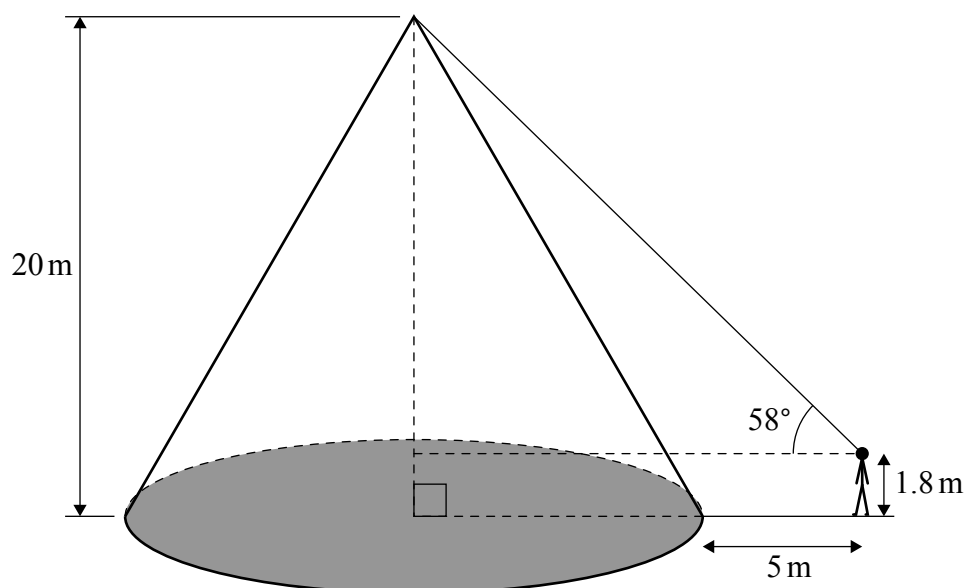


2. [Maximum mark: 5]

A monument is in the shape of a right cone with a vertical height of 20 metres. Oliver stands 5 metres from the base of the monument. His eye level is 1.8 metres above the ground and the angle of elevation from Oliver's eye level to the vertex of the cone is 58° , as shown on the following diagram.

diagram not to scale



- (a) Find the radius of the base of the cone. [3]
- (b) Find the volume of the monument. [2]

[illegible]